

Initiating Search

November 10, 2025, 5:07 PM

• Search:

CAS Lexicon Search:

Bromination - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (21,578)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,631)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

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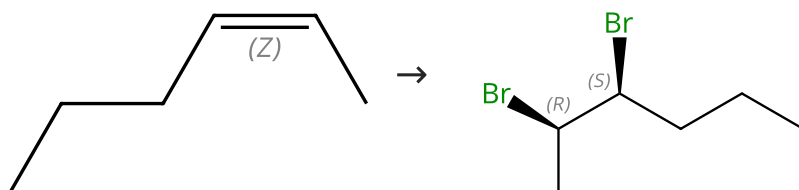
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Relative stereochemistry shown

 Suppliers (44)

 Supplier (1)

31-268-CAS-8512907

Steps: 1 Yield: 100%

Stereoselective Bromination Reactions Using Tridecylmethylphosphonium Tribromide in a "Stacked" Reactor

By: Ma, Kefeng; et al

Organic Letters (2008), 10(19), 4155-4158.

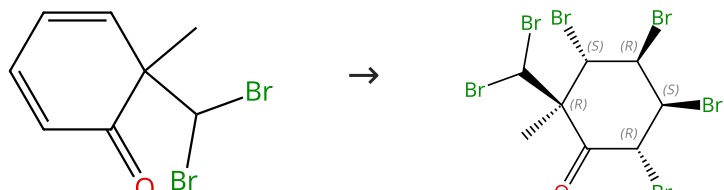
 1.1 **Reagents:** Bromine, Phosphonium, tris(decyl)methyl-, (tribromide) (1:1)

Solvents: Perfluorohexane, Hexadecane; 5 d, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

 Suppliers (5)

31-268-CAS-14390933

Steps: 1 Yield: 100%

Bromination of 6-dibromomethyl-6-methylcyclohexyl-2,4-dien-1-one

By: Gavrilova, G. V.; et al

Russian Journal of General Chemistry (2013), 83(10), 1844-1852.

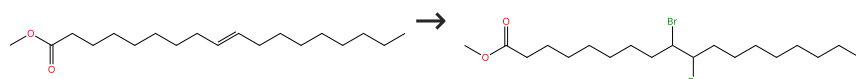
 1.1 **Reagents:** Bromine

Solvents: Chloroform-*d*; 2 h, rt

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (31)

 Suppliers (32)

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%

Double bond geometry shown

Suppliers (2)

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%

The reaction scheme shows the conversion of a bis-alkyne to a bis-alkene. The starting material is a bis-alkyne with two terminal vinyl groups and two internal alkyne groups, connected by a central chain. The product is a bis-alkene where the alkyne groups have been converted to alkenes, and the terminal vinyl groups have been converted to brominated alkenes. The reaction is labeled "Scheme 5 (1 Reaction)" and "Steps: 1 Yield: 100%".

• 2 Br⁻

• 2 Br⁻

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%

The reaction scheme shows the conversion of a bis-iminium salt to a bis-allyl bromide derivative. The starting material is a bis-iminium salt with two iminium groups connected by a central chain. The reagent is 2 equivalents of bromide ion (2 Br^-). The product is a bis-allyl bromide derivative where the iminium groups have been converted to allyl bromide groups.

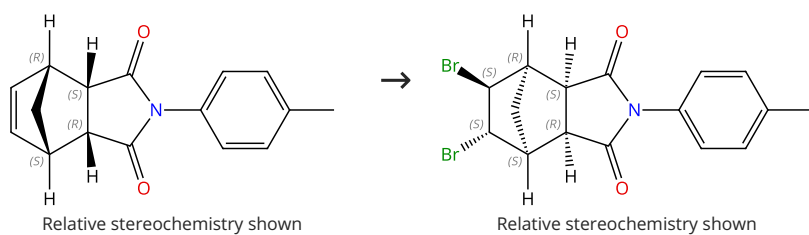
• 2 Br^-

• 2 Br^-

31-268-CAS-13987607	Steps: 1 Yield: 100%	Synthesis of 1,4-bis[dialkyl(2,3,4,5-tetrabromo-2-pentenyl) ammonio]-2-butene dibromides
1.1 Reagents: Bromine Solvents: Water		By: Gyul'nazaryan, A. Kh.; et al
1.2 Reagents: Acetone		Russian Journal of General Chemistry (Translation of Zhurnal Obshchei Khimii) (1999), 69(12), 1939-1941.
Experimental Protocols		

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (24)

Supplier (1)

31-268-CAS-16934129

Steps: 1 Yield: 100%

The quantum-chemical method of visualization of spatial structure of norbornenes

1.1 Reagents: Bromine

Solvents: Chloroform; -6 - -5 °C

By: Bagmanov, B. T.

Kimya Problemlari (2010), (2), 163-174.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

31-268-CAS-8640188

Steps: 1 Yield: 100%

High temperature bromination. Part 18: Bromination of benzonorbornadiene derivatives: Polybrominated benzonorbornenes and benzonorbornadienes

1.1 Reagents: Bromine

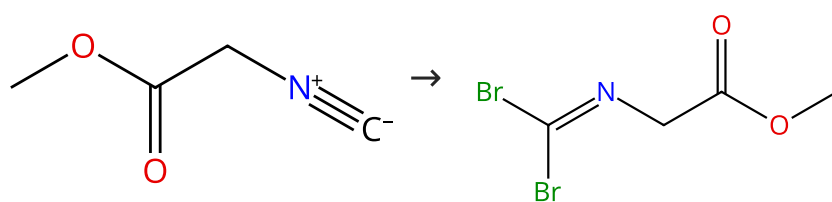
Solvents: Chloroform; 5 min, -10 °C

By: Dastan, Arif; et al

Tetrahedron (2005), 61(23), 5481-5488.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



31-268-CAS-550147

Steps: 1 Yield: 100%

Oxazole synthesis from isocyanides

1.1 Reagents: Bromine

Solvents: Acetonitrile; > 1 min, rt

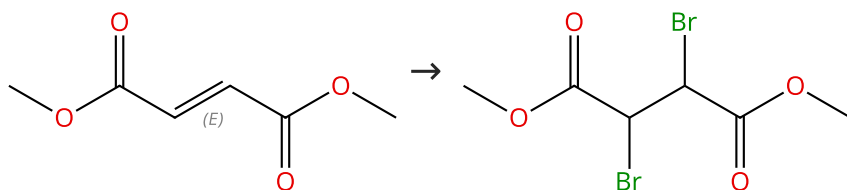
By: El Kaim, Laurent; et al

Synlett (2012), 23(9), 1361-1363.

Experimental Protocols

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Suppliers (111)

Suppliers (14)

31-614-CAS-38944681

Steps: 1 Yield: 100%

Synthesis of Methyl 3-Hydroxyisoxazole-5-Carboxylate via Bromination of Dimethyl Fumarate Under Photoflow Conditions and Its Safe Homologation into 3-(3-Methoxyisoxazol-5-yl) Propanoic Acid

1.1 Reagents: Bromine

Solvents: Dichloromethane; 25 - 35 °C; 30 min, 25 °C

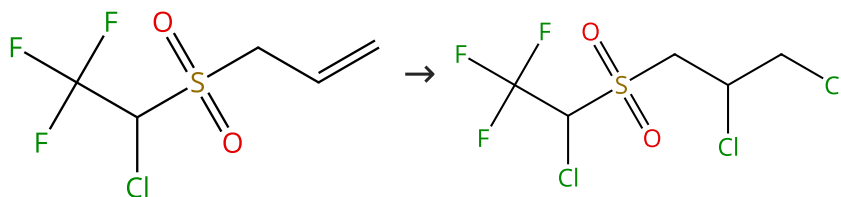
Experimental Protocols

By: Burki, Cedric; et al

Organic Process Research & Development (2024), 28(5), 1690-1703.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (2)

31-268-CAS-1055261

Steps: 1 Yield: 100%

An easy preparation of per- and poly-fluoroalkyl propenyl sulfones

1.1 Reagents: Chlorine

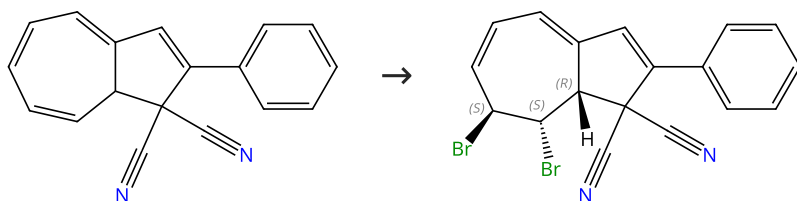
Solvents: Carbon tetrachloride; 1 h, rt

By: Dmowski, Wojciech; et al

Journal of Fluorine Chemistry (2005), 126(6), 877-882.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Relative stereochemistry shown
crystal structure

31-268-CAS-3142774

Steps: 1 Yield: 100%

New routes to functionalized dihydroazulene photoswitches

1.1 Reagents: Bromine

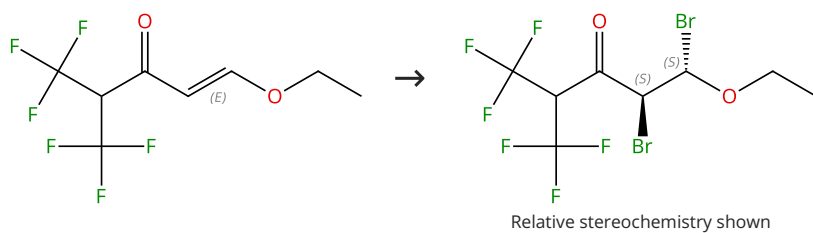
Solvents: Dichloromethane

By: Nielsen, Mogens Broendsted; et al

Pure and Applied Chemistry (2010), 82(4), 843-852.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

31-268-CAS-10285707

Steps: 1 Yield: 100%

Halogenation of β -alkoxyvinyl polyhaloalkyl ketones: a convenient route for the synthesis of α -chloro- or α -bromo- β -alkoxyvinyl polyhaloalkyl ketones

1.1 **Reagents:** Bromine
Solvents: Carbon tetrachloride

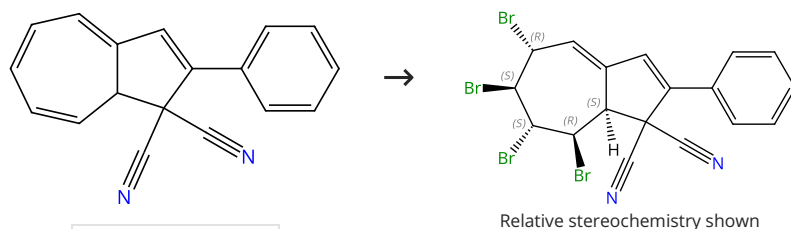
Experimental Protocols

By: Gerus, Igor I.; et al

Synthesis (2001), (3), 431-436.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

31-268-CAS-13624793

Steps: 1 Yield: 100%

New routes to functionalized dihydroazulene photoswitches

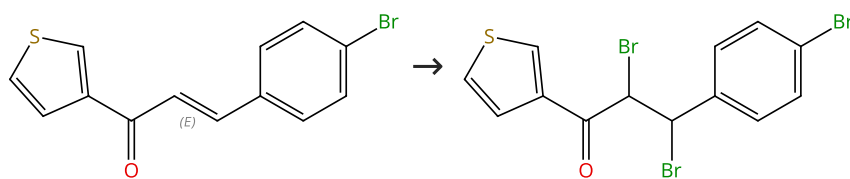
1.1 **Reagents:** Bromine
Solvents: Dichloromethane

By: Nielsen, Mogens Broendsted; et al

Pure and Applied Chemistry (2010), 82(4), 843-852.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

31-268-CAS-12215777

Steps: 1 Yield: 100%

Synthesis and characterization of α -bromo chalcone derivatives

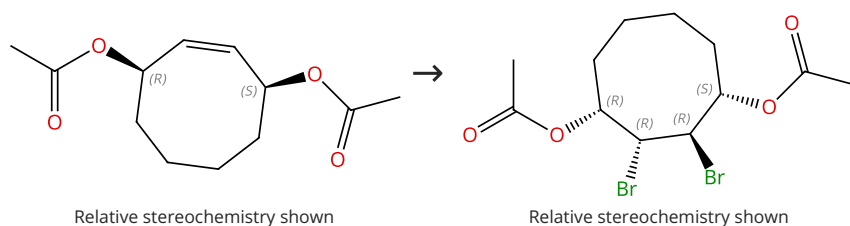
1.1 **Reagents:** Bromine
Solvents: Dichloromethane; 1 - 2 h, 25 °C

By: Budak, Yakup; et al

Chinese Journal of Chemistry (2009), 27(8), 1575-1581.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

31-268-CAS-18619461

Steps: 1 Yield: 100%

Synthesis of eight-membered aminocyclitol analogues

1.1 Reagents: Bromine

Solvents: Dichloromethane; -40 °C; 2 h, -40 °C

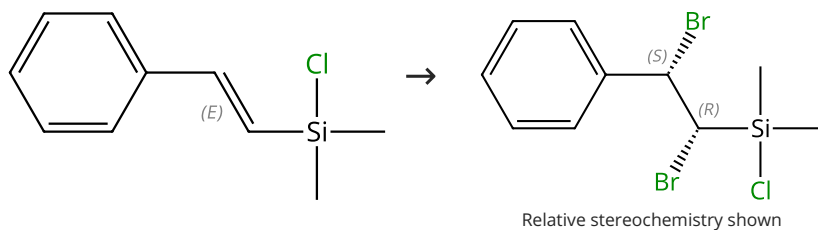
By: Zozik, Yunus; et al

Tetrahedron Letters (2017), 58(52), 4822-4826.

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

31-268-CAS-4141892

Steps: 1 Yield: 100%

An examination of the β -effect in an addition reaction with different ligands on silicon

1.1 Reagents: Bromine

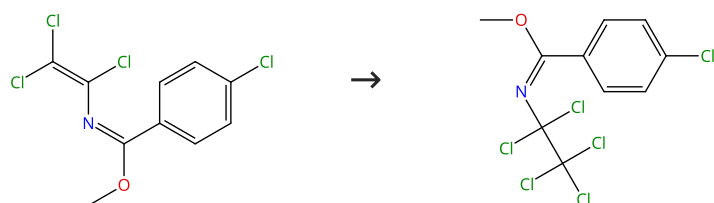
Solvents: Carbon disulfide

By: Brook, Michael A.; et al

Journal of the Chemical Society, Chemical Communications (1989), (14), 957-8.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Supplier (1)

31-268-CAS-10492193

Steps: 1 Yield: 100%

Halogenation of N-perchlorovinylimino ethers

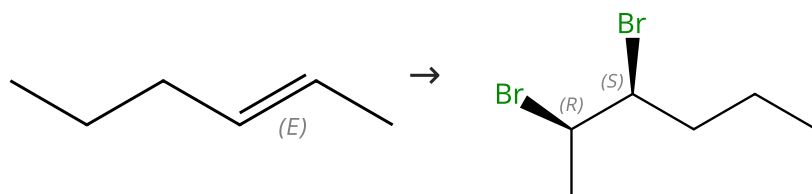
1.1 Reagents: Chlorine

By: Drach, B. S.; et al

Zhurnal Organicheskoi Khimii (1975), 11(9), 1913-17.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Relative stereochemistry shown

Suppliers (43)

Supplier (1)

31-268-CAS-9224319

Steps: 1 Yield: 100%

1.1 Reagents: Phosphonium, tris(decyl)methyl-, (tribromide) (1:1)

Solvents: Hexadecane; 10 d, rt

Experimental Protocols

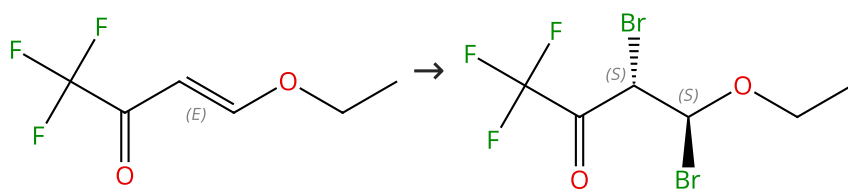
Stereoselective Bromination Reactions Using Tridecylmethylphosphonium Tribromide in a "Stacked" Reactor

By: Ma, Kefeng; et al

Organic Letters (2008), 10(19), 4155-4158.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Relative stereochemistry shown

Suppliers (58)

31-268-CAS-7869569

Steps: 1 Yield: 100%

1.1 Reagents: Bromine

Solvents: Carbon tetrachloride

Experimental Protocols

Halogenation of β -alkoxyvinyl polyhaloalkyl ketones: a convenient route for the synthesis of α -chloro- or α -bromo- β -alkoxyvinyl polyhaloalkyl ketones

By: Gerus, Igor I.; et al

Synthesis (2001), (3), 431-436.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Relative stereochemistry shown

31-268-CAS-3738281

Steps: 1 Yield: 100%

1.1 Reagents: Bromine

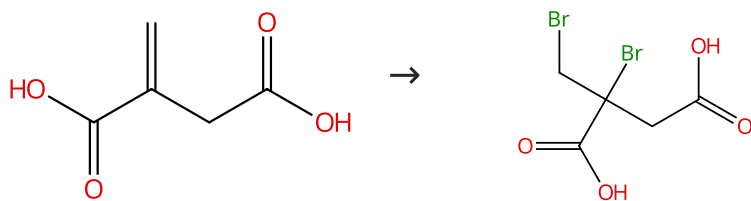
Solvents: Carbon tetrachloride; rt; rt $\rightarrow$ reflux; 5 min, reflux; 1 min, reflux; reflux $\rightarrow$ rt
Bromination of "endo-benzocyclobutanorbornene": Synthesis of "endo-11,12-dibromobenzocyclobutanorbornene": High-temperature bromination

By: Dastan, Arif; et al

Helvetica Chimica Acta (2002), 85(9), 2729-2739.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (6)

31-268-CAS-8160376

Steps: 1 Yield: 100%

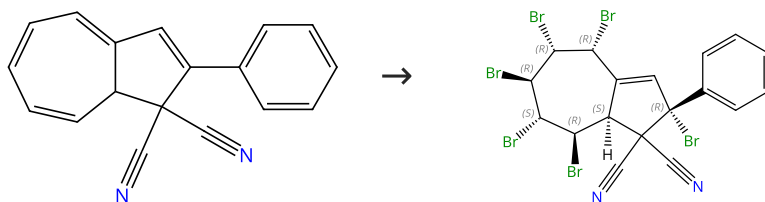
A new synthesis of 3-(bromomethyl)-2,5-furandione

- 1.1 **Reagents:** Bromine
Solvents: Carbon tetrachloride; reflux
- 1.2 **Reagents:** Sodium thiosulfate
Solvents: Water; cooled

By: Arfaoui, A.; et al
Journal de la Societe Chimique de Tunisie (2008), 10(1), 1-6.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Relative stereochemistry shown
crystal structure

31-268-CAS-233043

Steps: 1 Yield: 100%

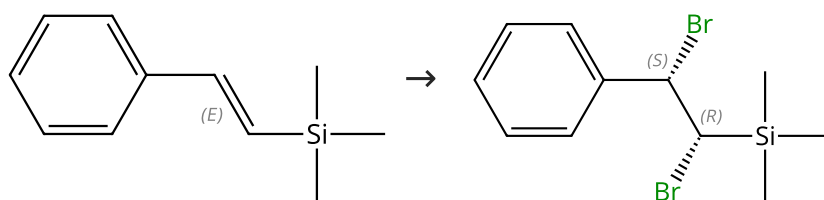
New routes to functionalized dihydroazulene photoswitches

- 1.1 **Reagents:** Bromine
Solvents: Dichloromethane

By: Nielsen, Mogens Broendsted; et al
Pure and Applied Chemistry (2010), 82(4), 843-852.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Relative stereochemistry shown

Suppliers (5)

31-268-CAS-2001606

Steps: 1 Yield: 100%

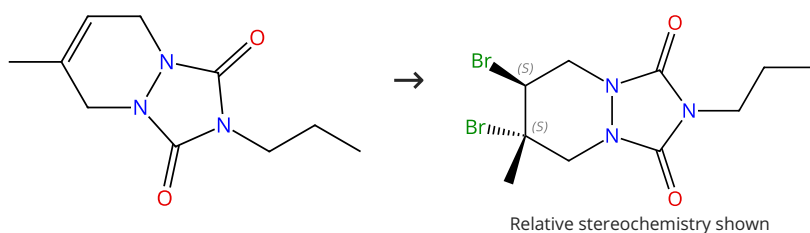
An examination of the β -effect in an addition reaction with different ligands on silicon

- 1.1 **Reagents:** Bromine
Solvents: Carbon disulfide

By: Brook, Michael A.; et al
Journal of the Chemical Society, Chemical Communications (1989), (14), 957-8.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



31-268-CAS-15101628

Steps: 1 Yield: 100%

Stereospecific vs stereoselective bromination of non-conjugated heterocyclic alkenes

1.1 Reagents: Bromine

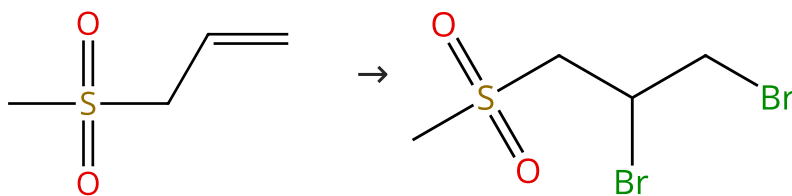
Solvents: Carbon tetrachloride

By: Mallakpour, Shadpour E.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (1995), 34B(4), 302-9.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (45)

Suppliers (2)

31-268-CAS-7850404

Steps: 1 Yield: 100%

An intramolecular bromonium to thiiranium ion rearrangement

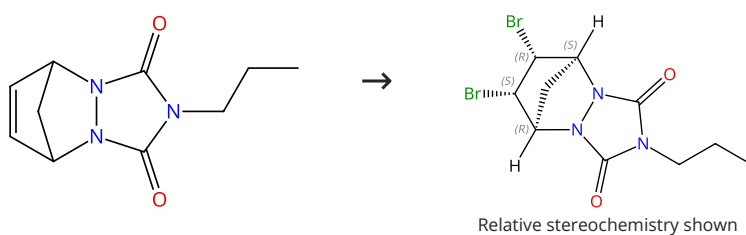
1.1 Reagents: Bromine

By: Bland, John M.; et al

Journal of Organic Chemistry (1983), 48(23), 4393-4.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



31-268-CAS-10572592

Steps: 1 Yield: 100%

Stereospecific vs stereoselective bromination of non-conjugated heterocyclic alkenes

1.1 Reagents: Bromine

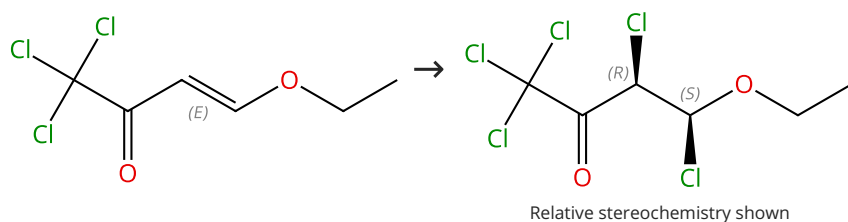
Solvents: Carbon tetrachloride

By: Mallakpour, Shadpour E.; et al

Indian Journal of Chemistry, Section B: Organic Chemistry Including Medicinal Chemistry (1995), 34B(4), 302-9.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (55)

31-268-CAS-9999188

Steps: 1 Yield: 100%

Halogenation of β -alkoxyvinyl polyhaloalkyl ketones: a convenient route for the synthesis of α -chloro- or α -bromo- β -alkoxyvinyl polyhaloalkyl ketones

1.1 **Reagents:** Chlorine
Solvents: Carbon tetrachloride

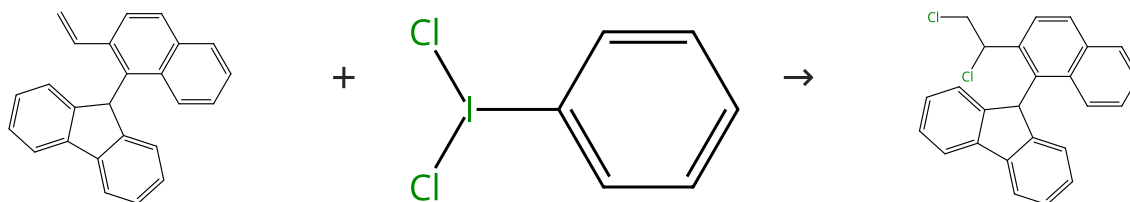
Experimental Protocols

By: Gerus, Igor I.; et al

Synthesis (2001), (3), 431-436.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (40)

31-268-CAS-18185430

Steps: 1 Yield: 100%

Reactivities of stable rotamers. XXIII. Some addition reactions toward the vinyl group in 9-(2-vinyl-1-naphthyl)fluorene rotamers

1.1 **Solvents:** Dichloromethane

By: Oki, Michinori; et al

Bulletin of the Chemical Society of Japan (1988), 61(12), 4303-8.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (104)

Suppliers (4)

31-268-CAS-11340660

Steps: 1 Yield: 100%

Stereoselective Bromination Reactions Using Tridecylmethylphosphonium Tribromide in a "Stacked" Reactor

1.1 **Reagents:** Phosphonium, tris(decyl)methyl-, (tribromide) (1:1)
Solvents: Hexadecane; 10 d, rt

By: Ma, Kefeng; et al

Experimental Protocols

Organic Letters (2008), 10(19), 4155-4158.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



31-268-CAS-10704365

Steps: 1 Yield: 100%

Bromination of endo- and exo-benzocyclobutenonornbornene derivatives: Neighbor group effect on bromination

1.1 Reagents: Bromine

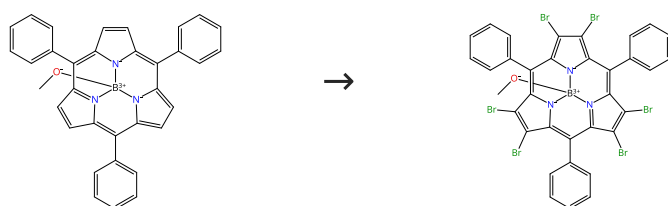
Solvents: Carbon tetrachloride; 10 min, 77 °C

By: Uzundumlu, Eren; et al

Journal of Chemical Research (2005), (6), 348-351.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



31-203-CAS-14752433

Steps: 1 Yield: 100%

Peripheral hexabromination, hexaphenylation, and hexaethylation of meso-aryl-substituted subporphyrins

1.1 Reagents: Bromine

Solvents: Chloroform; 2 h, rt

By: Tsurumaki, Eiji; et al

1.2 Reagents: Sodium thiosulfate

Solvents: Water; rt

Chemistry - A European Journal (2009), 15(1), 237-247.

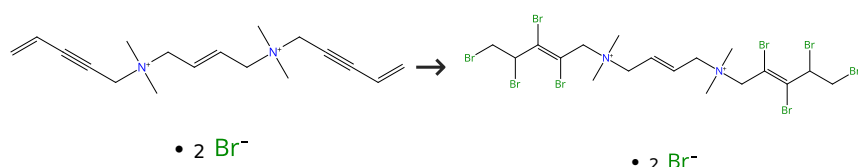
1.3 Reagents: Methanol

Solvents: Dichloromethane; rt → 50 °C; 10 min, 50 °C

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



31-268-CAS-9722182

Steps: 1 Yield: 100%

Synthesis of 1,4-bis[dialkyl(2,3,4,5-tetrabromo-2-pentenyl) ammonio]-2-butene dibromides

1.1 Reagents: Bromine

Solvents: Water

By: Gyul'nazaryan, A. Kh.; et al

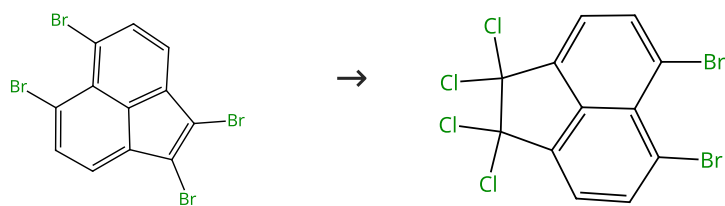
1.2 Reagents: Acetone

Russian Journal of General Chemistry (Translation of Zhurnal Obshchei Khimii) (1999), 69(12), 1939-1941.

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (3)

31-046-CAS-3822845

Steps: 1 Yield: 100%

Photochemical bromination of 5,6-dibromoacenaphthene

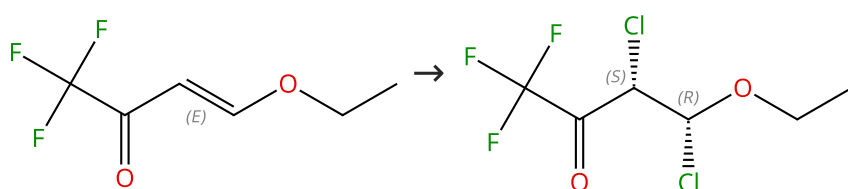
1.1 Reagents: Chlorine

By: Usachenko, V. G.; et al

Ukrainskii Khimicheskii Zhurnal (Russian Edition) (1975), 41(6), 663.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Relative stereochemistry shown

Suppliers (58)

31-268-CAS-5740714

Steps: 1 Yield: 100%

Halogenation of β -alkoxyvinyl polyhaloalkyl ketones: a convenient route for the synthesis of α -chloro- or α -bromo- β -alkoxyvinyl polyhaloalkyl ketones

1.1 Reagents: Chlorine

Solvents: Carbon tetrachloride

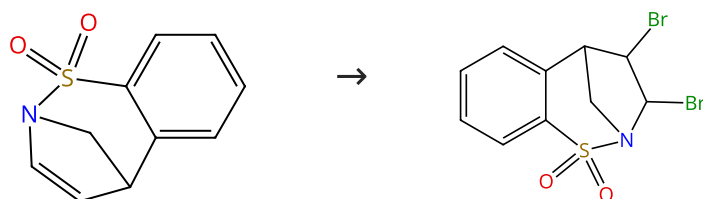
By: Gerus, Igor I.; et al

Experimental Protocols

Synthesis (2001), (3), 431-436.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

31-268-CAS-624347

Steps: 1 Yield: 100%

Brominative indolization of an unsaturated bridgehead sultam with extrusion of sulfur dioxide

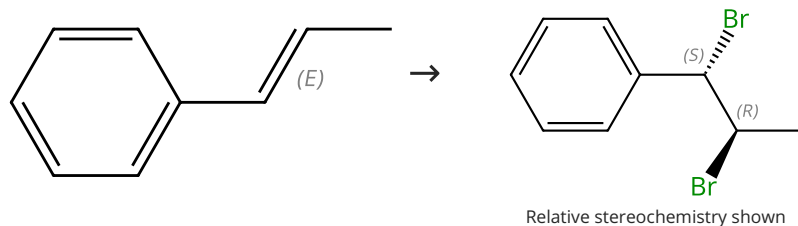
1.1 Reagents: Bromine; 14 h, rt

By: Preston, Adam J.; et al

Heterocycles (2006), 70, 41-44.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (59)

Suppliers (4)

31-268-CAS-13773636

Steps: 1 Yield: 100%

Concentration dependence of the steric course of the bromine addition to arylalkenes. The case of stilbenes

By: Bellucci, Giuseppe; et al

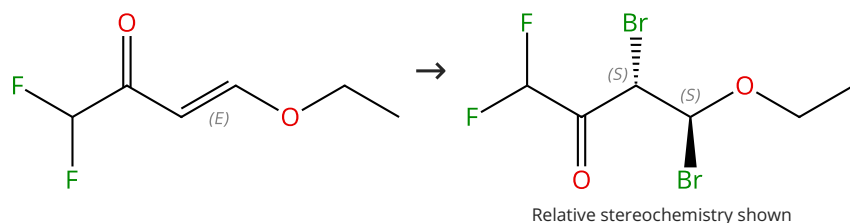
Journal of Organic Chemistry (1990), 55(13), 4094-8.

1.1 Reagents: Tetrabutylammonium tribromide

Solvents: 1,2-Dichloroethane

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (38)

31-268-CAS-3597807

Steps: 1 Yield: 100%

Halogenation of β -alkoxyvinyl polyhaloalkyl ketones: a convenient route for the synthesis of α -chloro- or α -bromo- β -alkoxyvinyl polyhaloalkyl ketones

By: Gerus, Igor I.; et al

Synthesis (2001), (3), 431-436.

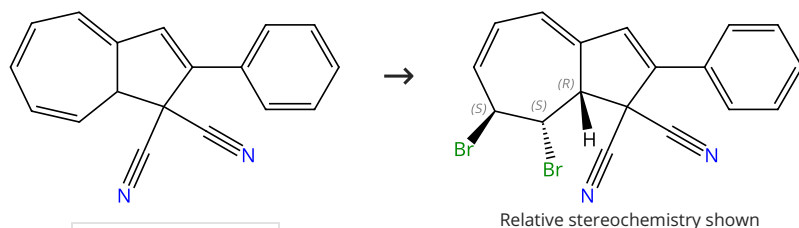
1.1 Reagents: Bromine

Solvents: Carbon tetrachloride

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

31-268-CAS-3460320

Steps: 1 Yield: 100%

A novel route to a bromo-cyano-substituted azulene and its exploitation in the construction of an acetylenic scaffold

By: Petersen, Michael Aaxman; et al

European Journal of Organic Chemistry (2007), (9), 1415-1418.

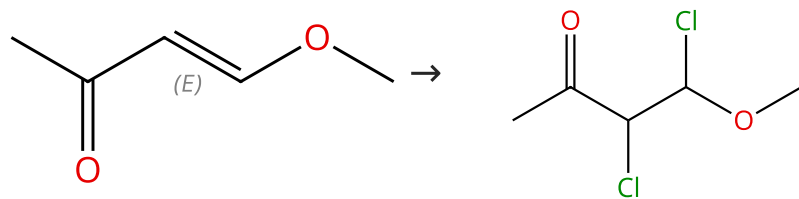
1.1 Reagents: Bromine

Solvents: Dichloromethane; 2 min, -78 °C; 1 h, -78 °C

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Suppliers (53)

Suppliers (4)

31-268-CAS-3511110

Steps: 1 Yield: 100%

Halogenation of β -alkoxyvinyl polyhaloalkyl ketones: a convenient route for the synthesis of α -chloro- or α -bromo- β -alkoxyvinyl polyhaloalkyl ketones

1.1 Reagents: Chlorine
Solvents: Chloroform-*d*

Experimental Protocols

By: Gerus, Igor I.; et al

Synthesis (2001), (3), 431-436.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 100%



31-203-CAS-14563493

Steps: 1 Yield: 100%

β -perfluoroalkylated meso-aryl-substituted subporphyrins: synthesis and properties

1.1 Reagents: Bromine
Solvents: Chloroform; rt

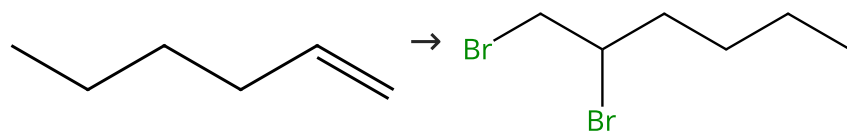
Experimental Protocols

By: Zhao, Shuai; et al

Synthesis (2014), 46(12), 1674-1688, 15 pp..

Scheme 42 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (76)

Suppliers (36)

31-268-CAS-19465581

Steps: 1 Yield: 100%

Practical regio- and stereoselective azidation and amination of terminal alkenes

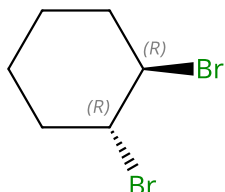
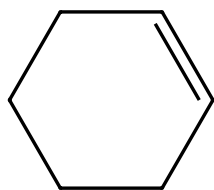
1.1 Reagents: Bromine
Solvents: Chloroform; 0 °C; 10 min, 0 °C

By: Ojo, Olatunji S.; et al

Organic & Biomolecular Chemistry (2018), 16(48), 9354-9358.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Suppliers (63)

Suppliers (63)

31-268-CAS-2836885

Steps: 1 Yield: 100%

Stereoselective Bromination Reactions Using Tridecylmethylphosphonium Tribromide in a "Stacked" Reactor

1.1 **Reagents:** Phosphonium, tris(decyl)methyl-, (tribromide) (1:1)
Solvents: Hexadecane; 12 d, rt

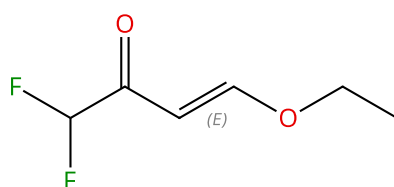
By: Ma, Kefeng; et al

Organic Letters (2008), 10(19), 4155-4158.

Experimental Protocols

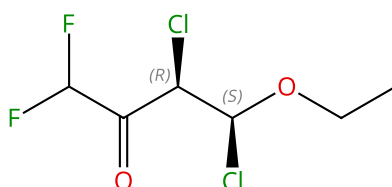
Scheme 44 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Suppliers (38)



Relative stereochemistry shown

31-268-CAS-1488880

Steps: 1 Yield: 100%

Halogenation of β -alkoxyvinyl polyhaloalkyl ketones: a convenient route for the synthesis of α -chloro- or α -bromo- β -alkoxyvinyl polyhaloalkyl ketones

1.1 **Reagents:** Chlorine
Solvents: Carbon tetrachloride

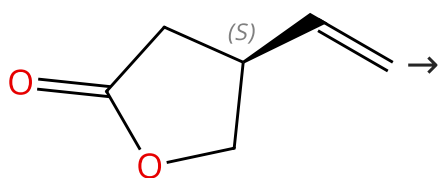
By: Gerus, Igor I.; et al

Synthesis (2001), (3), 431-436.

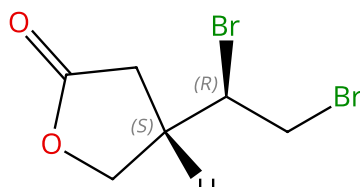
Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (+)

Suppliers (46)



Absolute stereochemistry shown

31-268-CAS-18332489

Steps: 1 Yield: 100%

Improved Synthesis of Racemate and Enantiomers of Taniguchi Lactone and Conversion of Their C-C Double Bonds into Triple Bonds

1.1 **Reagents:** Bromine
Solvents: Dichloromethane; 0 °C

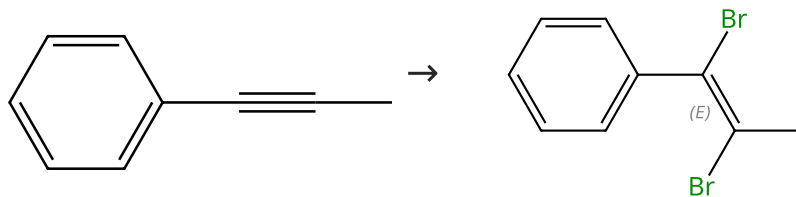
By: Malova Krizkova, Petra; et al

Synthesis (2018), 50(3), 651-657.

Experimental Protocols

Scheme 46 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

Double bond geometry shown

Suppliers (2)

31-268-CAS-7861559

Steps: 1 Yield: 100%

Vanadium-catalyzed oxidative bromination promoted by Bronsted acid or Lewis acid

By: Kikushima, Kotaro; et al

Tetrahedron (2010), 66(34), 6906-6911.

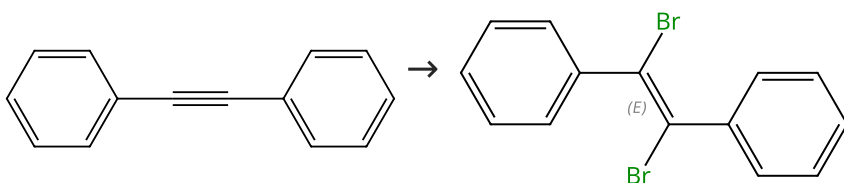
- 1.1 **Reagents:** Tetrabutylammonium bromide, Aluminum chloride, Oxygen
Catalysts: Ammonium vanadium oxide ((NH₄)VO₃)
Solvents: 1,4-Dioxane; 18 h, 80 °C

- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Scheme 47 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (79)

Double bond geometry shown

Suppliers (7)

31-268-CAS-12922336

Steps: 1 Yield: 100%

Bromide Ions and Methyltrioxorhenium as Cocatalysts for Hydrogen Peroxide Oxidations and Brominations

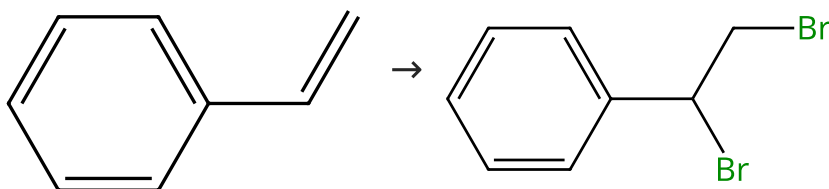
By: Espenson, James H.; et al

Journal of Organic Chemistry (1999), 64(4), 1191-1196.

- 1.1 **Reagents:** Sodium bromide, Hydrogen peroxide
Catalysts: Methyltrioxorhenium
Solvents: Acetic acid

Scheme 48 (3 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (115)

Suppliers (59)

31-268-CAS-18699289

Steps: 1 Yield: 100%

Copper(II) complexes of a heterotopic N-heterocyclic carbene ligand: Preparation and catalytic application

By: Aaron Lin, Shih-Chieh; et al

Journal of Organometallic Chemistry (2018), 859, 52-57.

- 1.1 **Reagents:** Trifluoroacetic acid, Lithium bromide, Oxygen
Catalysts: 2187493-16-7
Solvents: Acetonitrile; 20 h, 60 °C

Experimental Protocols

31-268-CAS-17940959		Steps: 1 Yield: 99%	Dibromination of alkenes with Li Br and H ₂ O ₂ under mild conditions By: Martins, Nayara Silva; et al New Journal of Chemistry (2018), 42(1), 161-167.
1.1	Reagents: Acetic acid, Lithium bromide Solvents: 1,4-Dioxane; 5 min, rt		
1.2	Reagents: Hydrogen peroxide Catalysts: Diphenyl ditelluride Solvents: 1,4-Dioxane; 1 h, rt		
Experimental Protocols			
31-268-CAS-6931785		Steps: 1 Yield: 99%	Perfluorohexane as a novel reaction medium for bromination reactions By: Pereira, Suzanne; et al Synthetic Communications (1995), 25(7), 1023-6.
1.1	Reagents: Bromine Solvents: Perfluorohexane		